

CSC311 ML CHALLENGE REPORT

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DATA

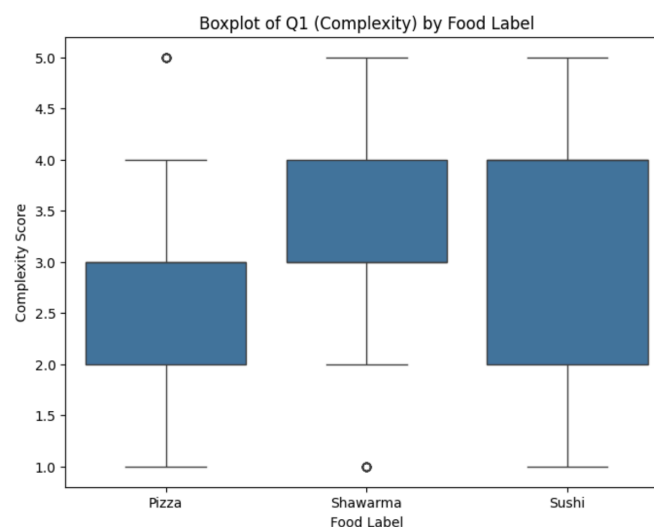
Exploration:

Q1 asks respondents to rate the complexity of making the food on a scale from 1 to 5 (with 1 being the simplest and 5 being the most complex). This feature gives us insight into the perceived difficulty or sophistication of each food type. When we group Q1 by the food label (e.g., Pizza, Shawarma, Sushi), we observe the following (sample summary):

Q1 Summary by Label:

	count	mean	std	min	25%	50%	75%	max
Label								
Pizza	548.0	2.875912	0.792518	1.0	2.0	3.0	3.0	5.0
Shawarma	548.0	3.226277	0.940595	1.0	3.0	3.0	4.0	5.0
Sushi	548.0	3.421533	1.210576	1.0	2.0	4.0	4.0	5.0

Key observations of the above data are that pizza has the lowest mean complexity (~2.88) and a smaller standard deviation (~0.79). This implies a relatively consistent perception that Pizza is simpler to make in general. Shawarma falls in between, with a mean ~3.23 and a moderate standard deviation (~0.94). Finally, Sushi shows the highest average complexity (~3.42) and the greatest variability (std ~1.21). This indicates that while many consider Sushi quite complex (median 4), there is also a substantial subset of respondents who gave it lower ratings, such as 1.



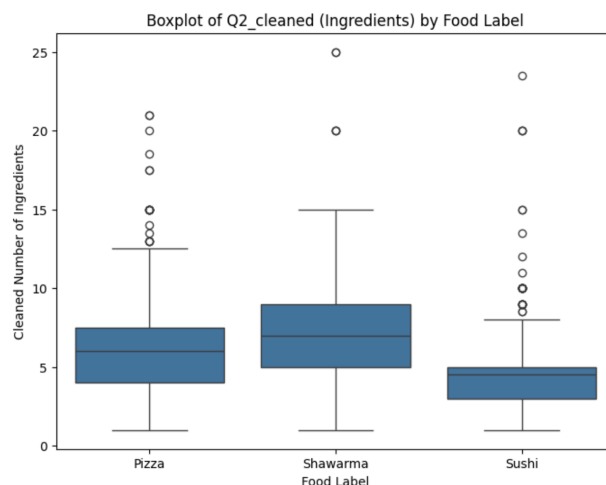
The above boxplot of Q1 (Complexity) by food label offers a visual comparison of how each dish's perceived complexity is distributed. Pizza appears to have the lowest median, sitting close to 3, and the narrowest spread, suggesting that most respondents rate its complexity

within a relatively tight range. The presence of a lower outlier around 1 indicates at least one individual who considers Pizza extremely simple, while a single upper outlier near 5 suggests a rare case where Pizza is viewed as highly complex. Shawarma, on the other hand, shows a slightly higher median, also near 3, but with a box that stretches toward the upper end of the scale, implying that more respondents find Shawarma at least moderately complex. In addition, the single lower outlier around 1 indicates that some see it as simple, but not many do so. Finally, sushi has a higher median, around 4, and a notably taller box, indicating greater variability in perceived complexity. Although many respondents rate Sushi near the top of the scale, a few place it as low as 1 or 2, resulting in a wide range of opinions. Overall, this boxplot highlights that while Pizza is generally considered simplest, and Sushi more complex, each label exhibits some overlap, reflecting the fact that personal experiences and definitions of “complexity” vary among respondents.

Q2 records the number of ingredients respondents expect the food item to contain. This feature has been “cleaned” (stored as Q2_cleaned) using a custom function that extracts numeric values (including handling ranges and textual numbers) or, in the absence of explicit numbers, counts ingredients using a predefined word bank. We observe the following sample summary:

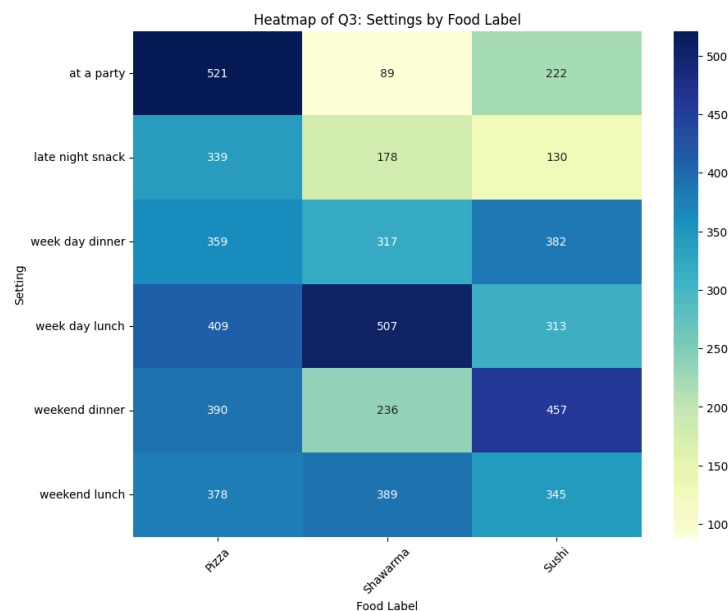
Q2 Summary by Label:								
	count	mean	std	min	25%	50%	75%	max
Label								
Pizza	546.0	6.256410	2.886890	1.0	4.0	6.0	7.5	21.0
Shawarma	545.0	7.252294	3.131775	1.0	5.0	7.0	9.0	25.0
Sushi	547.0	4.804388	2.453998	1.0	3.0	4.5	5.0	23.5

Key observations of the above data are that shawarma has the highest average ingredient count (~7.25), reflecting that it is a general consensus that it can include many different components. Pizza has an average ingredient count of 6.25, while sushi has the lowest count at around 4.80, along with the lowest variability in responses, as it has the lowest standard deviation amongst the labels at approximately 2.45. These results suggest that people see Shawarma as the most “ingredient-heavy” dish, Sushi as the simplest, and Pizza somewhere in between, with considerable overlap in each category due to differing interpretations of what constitutes an “ingredient.”



The boxplot for Q2_cleaned reveals that Shawarma tends to be viewed as having a richer array of ingredients, with its median ingredient count higher than that of Pizza and Sushi. Most Shawarma responses cluster toward a higher count, and there are several outliers suggesting that some respondents believe it can include a very large number of ingredients. Pizza, while having a moderate median, shows less spread overall, though there are a few cases where the ingredient count is notably high. Sushi appears to have the lowest typical ingredient count, with a lower median and a tighter distribution, despite a few occasional high responses. Ultimately, the visual differences indicate that Shawarma is perceived as the most ingredient-dense dish, Pizza falls in the middle, and Sushi is seen as simpler in terms of ingredients, although individual perceptions vary considerably.

Q3 asks respondents about the settings in which they would expect the specific food to be served. The following heatmap shows some patterns in how each food label is associated with various meal occasions.

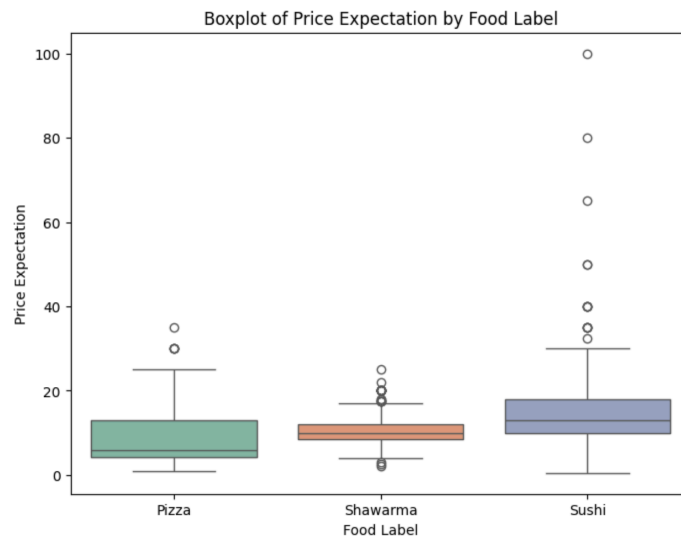


Pizza appears to correlate equally amongst all of the options for meal occasions; however, it is most strongly correlated with “at a party” (521), reflecting its appeal in social settings. Furthermore, shawarma shows its strongest association with “week day lunch” (507), suggesting it’s perceived as a quick midday option and it shows the least correlation with "at a party," suggesting it’s not typically viewed as a social dish. In comparison, sushi has the highest frequency of “weekend dinner” (457) responses, which implies that it is more often reserved for special occasions or a more upscale dining experience, and it is least associated with "late night snack." Overall, these patterns highlight that while Pizza is versatile in social contexts, Shawarma is favored for its convenience, and Sushi is considered for special occasions.

Q4 asks respondents to indicate how much they expect to pay for one serving of the food item. We observe the following sample summary:

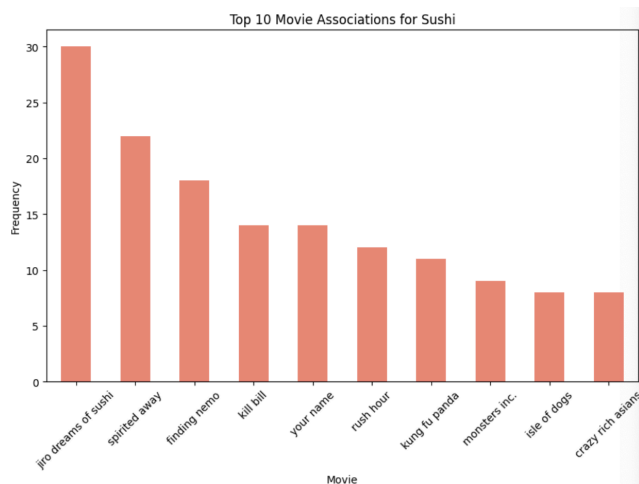
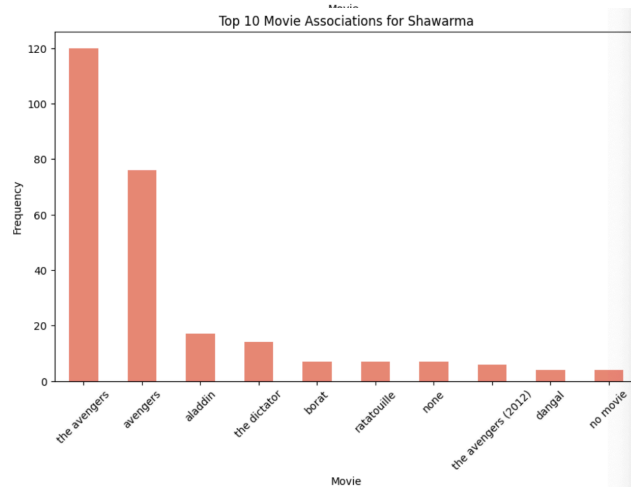
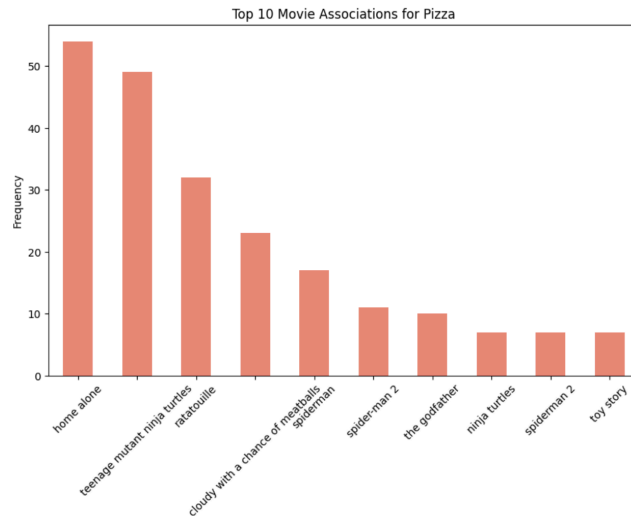
Q4 Summary by Label:								
	count	mean	std	min	25%	50%	75%	max
Label								
Pizza	546.0	8.825838	6.162098	1.0	4.26	6.0	13.0	35.0
Shawarma	544.0	10.708983	3.201408	2.0	8.50	10.0	12.0	25.0
Sushi	544.0	14.445132	8.872762	0.5	10.00	13.0	18.0	100.0

A key observation is that there are noticeable differences across the food labels. For instance, sushi exhibits the highest average price expectation (around \$14.45) with a broad spread (standard deviation of approximately \$8.87) and an extreme maximum of \$100. Moreover, shawarma has a mean price near \$10.71, with a tighter range and a maximum of about \$25, indicating a more consistent moderate cost expectation among respondents. Pizza shows the lowest mean price at roughly \$8.83, yet it still demonstrates considerable variability (standard deviation of about \$6.16) and a maximum around \$35, reflecting diverse opinions, possibly due to differences in one's interpretation of portion size (a whole pizza vs. a slice).



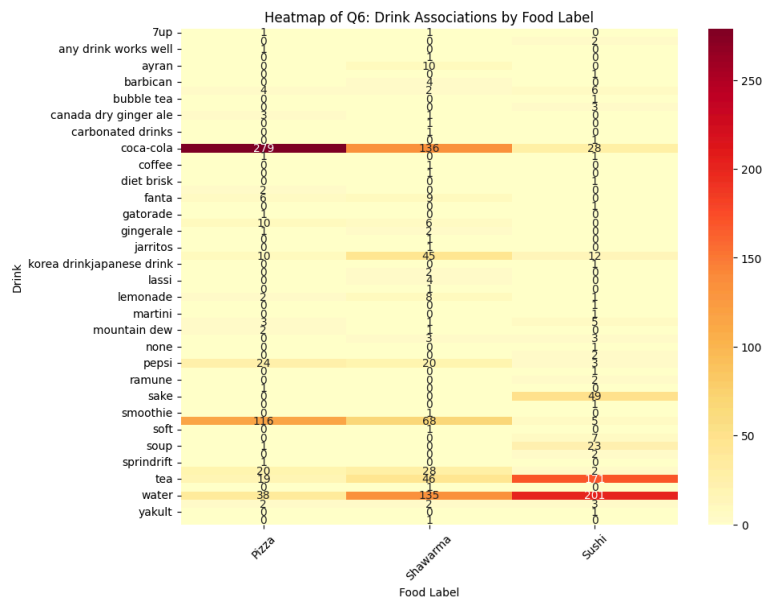
The above boxplot of Q4_Full_Price by food label visually confirms these findings. The boxplot results for shawarma indicate moderate and more consistent price expectations, as it has the smallest spread, indicating respondents generally agree on its cost. Sushi's box, however, not only sits at a higher level but also has more pronounced outliers, indicating that while most respondents expect a high price, there is substantial variability. Pizza has the lowest average price, but with a noticeable spread that implies differing opinions on what constitutes an "inexpensive" versus "expensive" pizza.

Q5 asks respondents to indicate which movie comes to mind when they think of the food item. The frequency counts for these movie features show how often each movie is associated with the food across all responses.



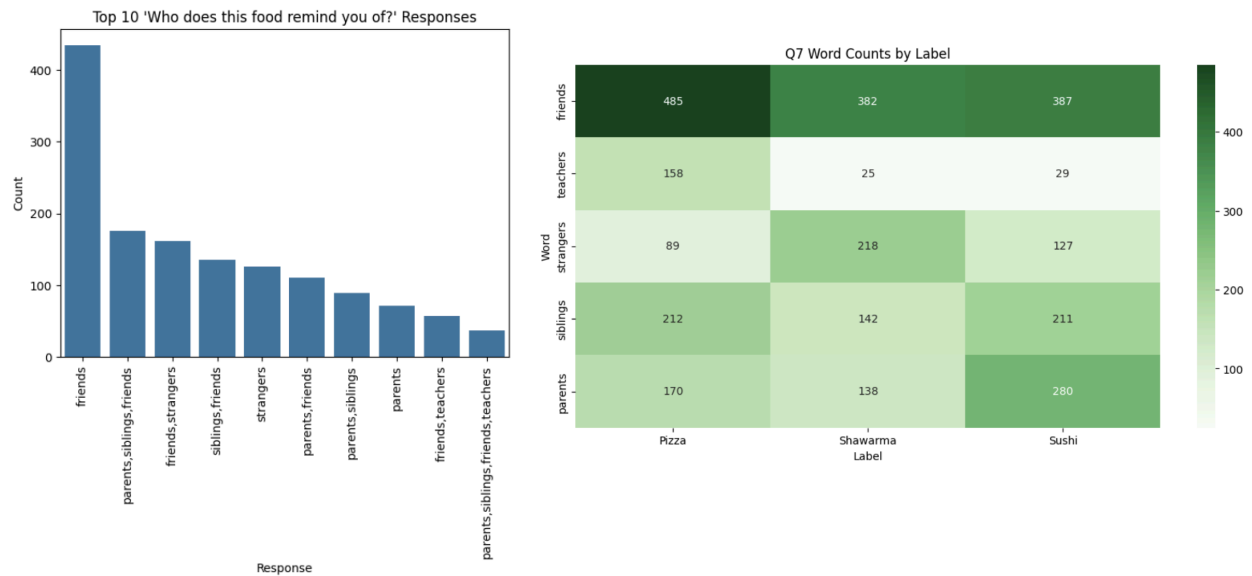
The movie associations vary distinctly across pizza, shawarma, and sushi, which reflects how each food item brings to mind different cultural or cinematic references in people’s minds. For example, for pizza, family-friendly or comedic films have higher frequencies, such as “Home Alone” and “Teenage Mutant Ninja Turtles.” By contrast, shawarma is overwhelmingly tied to “The Avengers,” due to the famous post-credit scene, suggesting a strong line between shawarma and the Marvel franchise. For sushi, the most frequently cited film is “Jiro Dreams of Sushi,” a documentary specifically about mastering sushi-making, which naturally resonates with respondents. In addition, other mentions, such as “Spirited Away,” and “Kill Bill” reflect Japanese cultural references. This data is important for our models because it captures nuanced cultural associations that may enhance the predictive power when classifying food items. Incorporating these movie associations as features can provide additional context that distinguishes between the food labels.

Q6 asks respondents about which drink they would pair with a specific food item. The following heatmap shows varying patterns in how each food label is associated with different drink options.



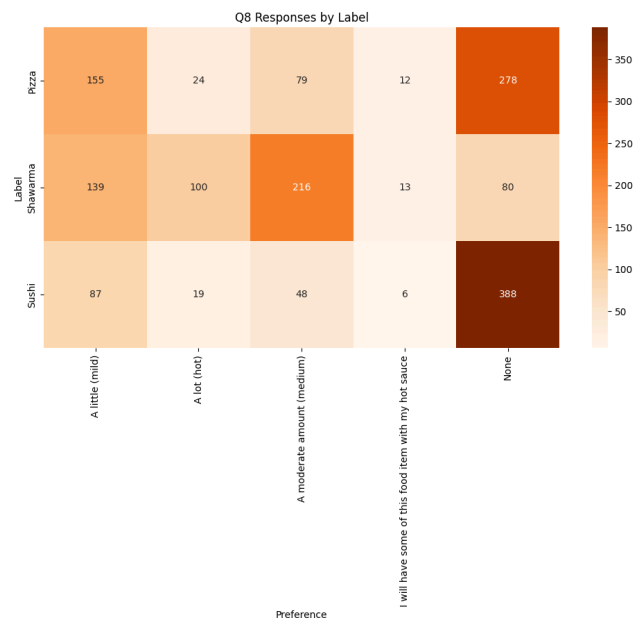
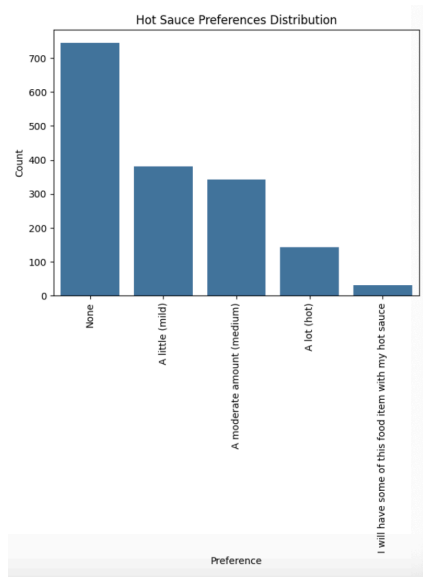
In this heatmap of Q6 (drink associations), each row represents a particular beverage while each column indicates a food label (Pizza, Shawarma, Sushi), with darker cells signifying a higher frequency of pairings. Pizza shows strong associations with soft drinks like soda and cola, aligning with its casual, social reputation. Shawarma is most commonly associated with water and cola, meanwhile, sushi is often linked with water, tea, or sake, which reflects more traditional or minimalist drink choices that match sushi’s taste profile.

Q7 asks respondents about people that come to mind when they think about a particular food. The following bar chart illustrates the top 10 responses, along with a heatmap that shows the frequencies with respect to each food label:



From the bar graph, it is evident that in general, “friends” stands out as the most frequent association by a large margin. This suggests that many people link these foods to social gatherings or shared experiences with their peers. Moreover, the heatmap above breaks down the frequency of each Q7 association by food label and we further see that “friends” remains the dominant response across all three categories, as indicated by its intense colour. However, the heatmap also shows label-specific nuances. For example, shawarma exhibits relatively higher frequencies for “strangers” compared to pizza and sushi, which may indicate that shawarma is more often linked to casual, everyday social environments. On the other hand, sushi appears to have a stronger association with “parents” and “siblings”, and pizza has a comparatively high frequency in association to “teachers.” In addition, it is also visible that shawarma has a lower association with siblings than the other two targets, so we could be able to use the absence of that in the response as an indicator that the target is not shawarma. Ultimately, while “friends” is prominent everywhere, the intensity and spread of other associations vary by label, emphasizing that each food item not only evokes a general social sentiment but also distinct relational contexts. This diversity in responses is valuable because it implies that the social or cultural associations captured in Q7 can contribute nuanced information for our models, potentially improving classification accuracy.

Q8 asks respondents about their hot sauce preferences for each food. The following bar chart illustrates the general frequencies for each hot sauce preference, along with a heatmap that shows the frequencies with respect to each food label:



In regards to the bar graph, “none” and “a little (mild)” stand out as the most frequent choices, suggesting that while many respondents enjoy some spice, a significant portion either prefers no hot sauce or is cautious. Fewer people opt for a “moderate amount (medium)” or “a lot (hot),” indicating that truly spicy preferences are less common in the overall sample. Furthermore, the heatmap on the right breaks these preferences down by food label. From the heatmap, we can see that pizza is most frequently associated with “none” or “a little (mild)”, suggesting that hot sauce is not a prominent component for many consumers. In contrast, shawarma tends to have a higher frequency of moderate to high hot sauce preferences, aligning with the idea that spicy flavors are more integral to its taste profile. Sushi is most strongly associated with “none”, and has less responses than pizza for “a little” (mild), even though it is its second most frequent option.

Feature Selection:

Although our exploration of Q1 revealed that Pizza consistently scores lower (mean ~2.88) compared to Sushi (mean ~3.42) and Shawarma (mean ~3.23), the real strength of Q1 lies in its ability to capture the perceived effort or sophistication in food preparation. This distinction not only helps differentiate between dishes but also correlates with other features such as ingredient count (Q2) and price expectation (Q4). For instance, pizza is consistently viewed as the simplest to prepare, evidenced by its lower complexity rating, an average of 6.25 ingredients (Q2), and thus correlates with the lowest price per serving (Q4). This correlation among features underscores that Q1 not only distinguishes the ease of preparation but also reinforces related culinary perceptions, making it a valuable input for our model. Moreover, the lower complexity ratings for Pizza align with its social and accessible nature (as also indicated by its strong association with “friends” and “at a party” in Q7 and Q3), making it a reliable indicator of a more

casual dish. In contrast, the higher and more variable ratings for Sushi signal a more premium or elaborate dish, which may influence higher price expectations.

Q2 was also selected as an input feature and quantifies the number of ingredients expected in each food item. Our data show that shawarma has the highest average (around 7.25), followed by pizza (6.25) and then sushi (4.80), reflecting a clear differentiation in perception. This feature is important to include in our model as it complements Q1; dishes perceived as more complex (like sushi) tend to have fewer ingredients, while those with moderate complexity (such as shawarma) are seen as more ingredient-dense. The correlation between Q1 and Q2 helps the model capture not just how difficult a dish is to prepare, but also how rich or layered its composition might be, providing a nuanced view of each food type.

Q3 captures the context in which respondents expect the food to be served. For instance, the strong association of pizza with “at a party” complements Q7, where “friends” is dominant, emphasizing pizza’s role as a social food. Meanwhile, shawarma’s link with “week day lunch” highlights its convenience and suitability for a quick meal, and sushi’s association with “weekend dinner” points to a more upscale or special occasion context. These are clear distinctions in our data, which we can use to predict the labels of the food item described, and thus allows our models to incorporate situational and cultural factors, which can significantly enhance predictive accuracy.

Q4 captures respondents’ price expectations for a serving of the food. The analysis shows that sushi has the highest average price (around \$14.45) with wide variability, while pizza is perceived as the least expensive (mean ~\$8.83) despite some spread. This pricing information not only reflects economic perceptions but also ties into the perceived quality and complexity of the food. For example, the higher price expectation for sushi aligns with its higher complexity rating (Q1, indicating a premium product, whereas pizza’s lower cost corresponds to its casual, everyday appeal as evidenced in its social associations. Including Q4 in the model helps the algorithm learn associations between economic perception and the food label. For instance, if a new observation includes a high price expectation, it increases the likelihood of the item being sushi rather than pizza. Similarly, a lower price estimate may steer the model toward predicting pizza, especially when supported by low complexity and high social association features.

Q5 was chosen as an input feature because it captures consistent cultural associations that enhance model accuracy. For example, from our previous data analysis, pizza was strongly linked to family-friendly or comedic films, shawarma was overwhelmingly linked to “The Avengers”, and sushi to culturally specific documentaries like “Jiro Dreams of Sushi.” By incorporating Q5, the model gains a semantic layer that complements quantitative features, improving its ability to accurately distinguish between food types.

The patterns observed in Q6 are distinct for each food type: for example, pizza is strongly paired with soft drinks such as soda and cola, reinforcing its casual character, while shawarma, which is commonly associated with water and cola, suggests a preference for everyday beverages, and sushi is predominantly linked with water, tea, or sake, reflecting more traditional

and minimalist drink choices. These differences are not only statistically significant but also align with the broader social and culinary narratives captured in Q3 and Q7. By including Q6 in our model, we enhance predictive accuracy because the unique beverage pairings serve as discriminative signals that help differentiate between food types. For instance, if a dish is associated with soft drinks and has a strong social context (as indicated by Q7), the model can more confidently classify it as pizza.

Q7 was chosen as an input feature as it highlights the social context in which a food is consumed by asking whom the food reminds respondents of. For example, pizza is most strongly associated with “friends,” highlighting its social and shareable nature, shawarma is more often linked with “strangers,” suggesting it is viewed as a casual and everyday option, and sushi is frequently associated with “parents,” indicating a more traditional or familial setting. These distinct cultural signals add valuable qualitative data that, when encoded appropriately, improve the model’s ability to differentiate food labels and enhance overall prediction accuracy.

The above data analysis for Q8 shows that pizza is most frequently paired with “none” or “a little (mild),” while Shawarma tends to attract stronger spice preferences and sushi is predominantly linked with “none.” These distinct flavor profiles, when combined with answers from Q3 (meal occasion) and Q7 (social associations), allow for better classification. For example, a dish that is associated with mild hot sauce preferences, a social setting of “at a party,” and strong ties to “friends” is more likely to be pizza, whereas a food with higher spice preferences, a setting of “week day lunch,” and associations with “strangers” would lean toward shawarma. Thus, we chose Q8 as an input feature as it is a crucial feature that enhances prediction accuracy by adding some layer of nuance that differentiates the food labels.

After data cleaning, for the Random Forest model, we decided to not use any form of dimensionality reduction since Decision Trees and Random Forest models generally do not suffer from the curse of dimensionality. Trees handle high-dimensional data well because they split features independently.

Data representation

For Q1, the responses for measuring the complexity of making the food on a scale from 1 to 5, are directly represented as a single numeric feature. This value is extracted from the original responses (which are already in numeric form) and missing values are replaced with the median to ensure consistency. This approach is effective for our Naive Bayes, Softmax Regression, and Random Forest models, as they can naturally handle numeric inputs and benefit from having precise, continuous values.

Since our Naive Bayes, Softmax Regression, and Random Forest models can handle numeric inputs, for Q2, which captures the expected number of ingredients, a custom cleaning function is applied that first attempts to extract numeric values from the free-form text (including handling ranges and textual numbers) and, if none are found, counts the tokens matching a predefined word bank. This process converts Q2 into a continuous numeric feature that reflects

respondents' expectations of ingredient richness. Representing Q2 as a continuous numeric feature helps show inherent differences in ingredient counts, complementing Q1's measure of complexity.

For Question 3 and 6, each of the columns are extracted from the original data sheet and then cleaned to have proper mapping associations as well as remove any "unavailable data" or data that has errors such as punctuation, trailing white space etc. This was then converted into a bag of words form, a vector that is used to indicate any word with a 1 or 0. It was chosen due to it being easier to standardize and normalize the data in a way so that the exploration and analysis becomes easier.

For Question 4, we found all of the numerical values within the text and used the maximum numerical value as the data point label. We used the max value of the text because we concluded it would be easier to classify the food type depending on its largest potential price, according to the data exploration for Q4 where the variation and mean price varied per food group (sushi vs pizza). For missing values, we populated those with the median value of the feature.

Question 5 was represented using bag-of-words, where we use the full movie name as the feature instead of the words because the redundant words that occur in multiple movies across different labels would make the prediction to be inaccurate by our judgment. This is a bag-of-words approach because data points containing any word in the titles will become 1. Nan values were filled with empty strings.

We decided to represent Question 7 as a bag of words, as it made more sense to split each word into its own category, and as each response could have multiple categories contained in it, we needed a representation that could indicate the presence of multiple categories, which is why we used a bag of words.

We decided to represent Question 8 as a one-hot vector, as each response only contained one category, and we needed to convert the words into a format our models could interpret.

Data splitting:

For the Naive Bayes model and the Random Forest Model, we experimented with multiple ways of splitting the dataset, by testing random splittings of proportions 80:20 (train:validation) and 70:30. We also tried splitting by the natural split that was in the csv file (where the categories repeated again), using a (921:723) split of the data. As the natural split in the csv file felt too even, we chose to use the 70:30 split.

In our softmax regression model, we split the complete dataset of 1,644 samples into training and test sets by using a natural split: the 723 samples were designated as the test set and the remaining 921 samples as the training set. Although this split was based on row order rather

than random assignment, it maintained a balanced distribution of labels (pizza, shawarma and sushi) across both sets.

For Naive Bayes and Random Forest, when doing grid search, we chose to use 5-fold cross validation. This would allow us to make sure that the accuracy of the chosen value for the hyperparameters (alpha, max_depth etc) was accurate and not just a lucky split of the dataset, ensuring that we would be able to choose the best overall hyperparameter to use. For the softmax model, this approach allowed us to select the best regularization parameter (C) based on performance across multiple data partitions, ensuring that our final model generalizes well to unseen data.

Model

Naive Bayes

The naive bayes model was trained using the multinomialNB model provided by numpy. This is a variation of the model learned in class, which used binomial targets (2 targets), as we have 3 different targets. We chose to use this model as it has high resistance to problems with high dimensional inputs, and as many of our features would be provided as massive vectors, we needed a model that could use these features without issues. Furthermore, as many of the responses used a bag of words for encoding and the responses were mostly not in sentence structure format (the words are independent of each other), the naive bayes model would be able to perform well on this data as it assumes the features are conditionally independent.

Random Forest

The second model we decided to use was Random Forest (using RandomForestClassifier). This model uses an ensemble that combines the predictions of multiple decision trees to classify data points. This improves accuracy, reduces variance and overfitting. Since the dataset has a lot of features post data cleaning (like several one hot encoded variables), this model is also a good choice since it handles high dimensional data well. In addition, it automatically selects important features, reducing the need for manual feature selection and handles categorical and numerical features.

Softmax Regression

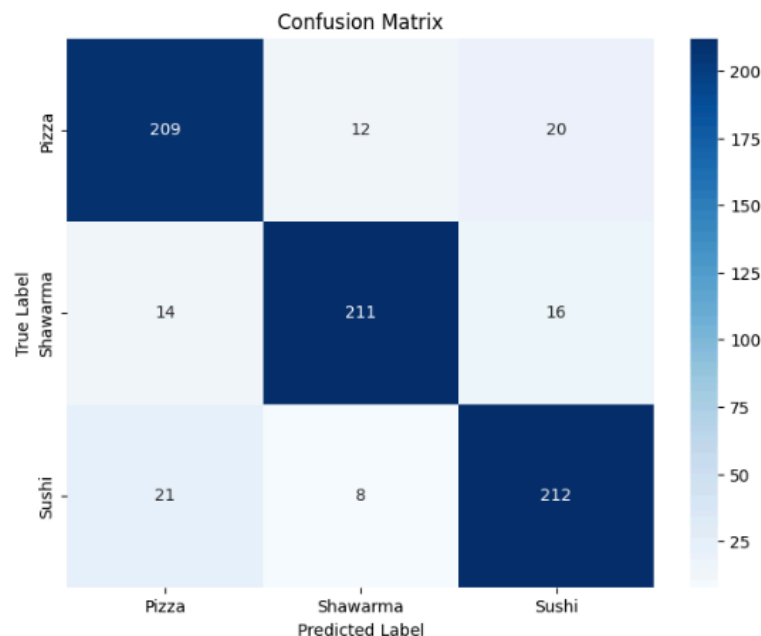
Softmax regression was chosen because it naturally handles multi-class classification by converting raw model outputs (logits) into a normalized probability distribution over classes. This characteristic is particularly valuable for our task, where we need to predict food labels across multiple categories. The model's probabilistic output, where each prediction is accompanied by a confidence score that sums to one, provides a clear interpretation of how likely a food item is to belong to each class. Moreover, softmax regression effectively leverages our continuous input features, such as complexity (Q1), ingredient count (Q2), and price expectation (Q4), to capture subtle differences between food types, thereby enhancing overall prediction accuracy.

Model choice and hyperparameters:

To compare our models, we primarily used validation accuracy and the confusion matrix. This is because validation accuracy provides a reliable estimate of how well the model generalizes to unseen data, helping us avoid overfitting and compare model performance fairly across different algorithms. In addition, the confusion matrix offers a more detailed breakdown by showing how predictions are distributed across each class. This allowed us to pinpoint specific misclassifications and assess whether the model was biased toward or struggling with any particular class.

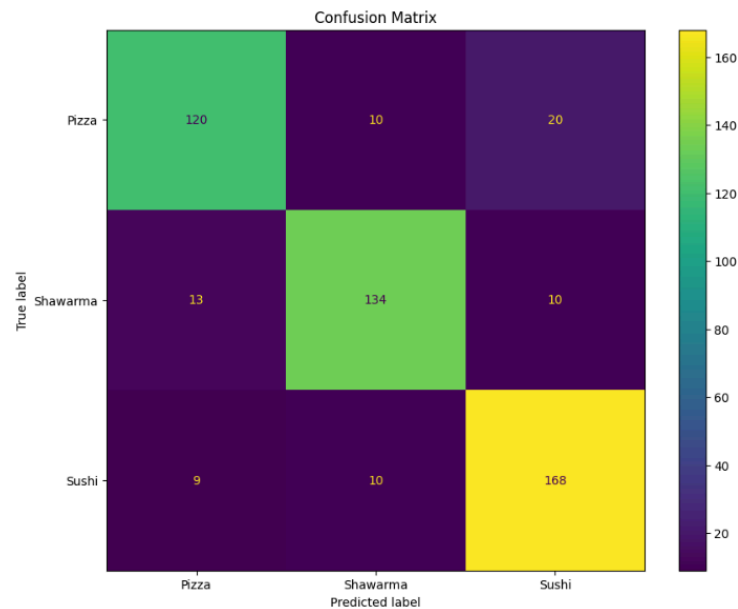
For the Softmax model, a problem that can happen when training a model is overfitting (in other words, when the model becomes overconfident and fits the data a little too well). With complex models, the model can fixate and memorize the training data and fits over any noise. This causes the model not to generalize well. To combat this, we were able to tune the model using regularization by allowing the model to focus on features that are more important and penalizing others. Regularization penalizes large weights, with small weights there are smaller fluctuations and less noise. This turning helped improve the model's ability to generalize while maintaining high predictive performance. As a result, the model achieved a training accuracy of 0.9251 and a validation accuracy of 0.88.

The confusion matrix, shown below, demonstrates that the softmax model performs consistently well across the three classes. For instance, pizza, shawarma and sushi all have over 200 correct predictions, with relatively low misclassifications.



For the Naive Bayes model, we tuned alpha, which is a smoothing parameter that handles how zero probabilities are dealt with (i.e., when a word is not seen in the training set for a given class). It does this by adding it to the counts for the probabilities of a word in a class. After performing a grid search, we found that $\alpha = 0.1$ was the best value we got out of the values

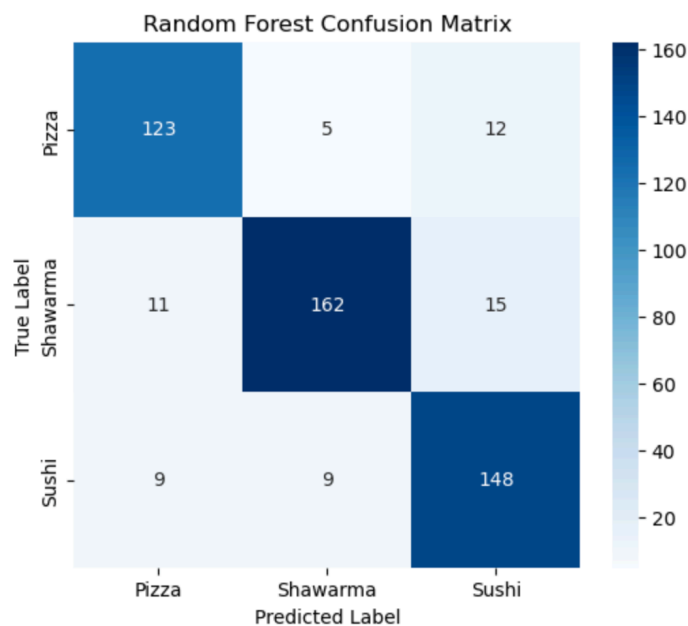
we searched. The metrics we used to find the best hyperparameter were accuracy, ROC (receiver operator) curves, and a confusion matrix. The accuracy gave us a general overview of how the hyperparameter did, while the other two provided insights into which specific targets the model did not perform well on. Something we noticed before hyperparameter tuning was that pizza kept getting confused for sushi, which we could see in both the ROC curve for pizza and the confusion matrix. We specifically tried to reduce this error while not increasing the error of the other predictions, and the value $\alpha = 0.1$ reduced this error the most while keeping the other predictions around the same/more accurate. As a result, the model achieved a training accuracy of 0.9504, showing strong fit to the training data and a validation accuracy of 0.8543.



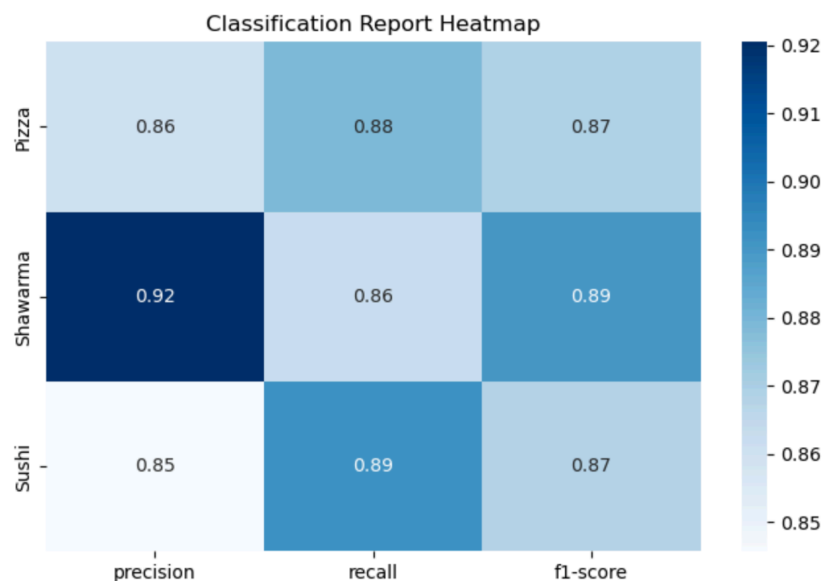
For the Random Forest model, the hyperparameters we tuned included `n_estimators` (number of trees), `max_depth` (tree depth), `min_samples_split` (minimum samples required to split a node), and `min_samples_leaf` (minimum samples per leaf). The metrics we used to find the best hyperparameter were validation accuracy, classification report and confusion matrix. The validation accuracy tells us how the model parameters perform on unseen data, the classification report tells us whether the model favors certain classes (e.g., if some classes have low recall) or if the dataset is imbalanced. The confusion matrix pinpoints specific misclassification to help diagnose bias towards a particular class.

We tested different values, ranging from 50 to 200 trees, varying depths up to 30 or unlimited, and adjusted split size ranging from 2 to 20. The best combination of hyperparameters was `{'max_depth': 10, 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_estimators': 200}`, which resulted in a training accuracy of 0.889 and a validation accuracy of 0.876, indicating strong generalization. However, the model struggles a bit with misclassifications, particularly for Shawarma, where 15 data points were predicted as Sushi. This is reflected in the classification report, where Shawarma has the lowest recall (0.86), meaning a good portion of actual

Shawarma samples were misclassified. However, overall the confusion matrix shows that the model did a good job of predicting the correct label.



Despite this, the model achieves an overall accuracy of 0.876, demonstrating strong predictive power. The classification report confirmed that the model maintained good precision and recall.



The model we ultimately decided to use was the **Softmax** regression model. Among all models tested, softmax achieved the highest validation accuracy (0.88), indicating better generalization to unseen data in comparison to the other labels. This was further supported by the confusion matrix, where all three classes of food labels had over 200 correct predictions, demonstrating

balanced performance and the lowest rate of misclassification (the ratio of correct vs. incorrect predictions). In contrast, both the Naive Bayes and Random Forest models showed greater class-specific misclassifications, and lower validation accuracies (0.854 for Naive Bayes, 0.876 for Random Forest).

Moreover, Softmax was also the most suitable model for this multi-class classification problem, where each input needed to be confidently assigned to a single class. Along with the ease of interpreting the data since it produces a prediction/confidence score that sums to one (probability distribution). The ease in tuning the hyperparameters, in this case regularization to reduce overfitting, was also a benefit in choosing softmax.

Prediction:

We estimate to get 85% on the test data set. This estimate is based on our model's 88% accuracy on the validation set, which provides a reliable indicator of how well the model generalized to unseen data. However, given that the responses for the questions could change slightly for the questions using a bag of words representation for the data, causing some responses to have the entire vector set to 0 in the case where all words are new, we think that the accuracy of the model will drop slightly on the test set since there is a chance that the test set may contain entirely new words not seen during training.

Workload Distribution

Mahek: I developed the softmax regression model and handled data cleaning for Q1 and Q2, while also writing and conducting comprehensive exploratory data analysis across Q1-Q8 in the report.

Hai: I trained the naive bayes model and wrote the prediction script, and handled data cleaning and initial data analysis for Q7 and Q8. I also wrote parts of the report related to the naive bayes model.

Nimrat Grewal: I developed/wrote the code for the random forest model, handled data cleaning for Q4 and Q5. I also wrote all parts of the report that related to the Random Forest Model and the Data Splitting section concerning Q4 and Q5.

Abhishek: Helped with the softmax model and added the hyperparameter tuning for the regularization C. Did the cleaning and data splitting for questions 3 and 6, as well as the reporting related to those questions and some of softmax (hyperparams).